

The named, stage-by-stage version — who to meet, what to build, how to get funded

The Inside Track.

An operator's map for a pharmaceutical-chemistry student who wants to build something — and the one frontier that unites everything he's drawn to.

Prepared for **Marcus** · University of Guelph, Biological & Pharmaceutical Chemistry · the entrepreneur's edition

The thesis: one frontier unites everything

Where his nuclear pull, his pharmaceutical-chemistry degree, his entrepreneurial drive — and his own story — meet

Until now I split him into a "nuclear track" and a "pharma track." That was a mistake. There is a single field that is **both at once**, it is the hottest frontier in Canadian biotech, and the role model for it is a chemistry professor who built a multi-billion-dollar cancer company in Ontario.

RADIOPHARMACEUTICALS — THE UNIFYING FIELD

Radiopharmaceuticals (targeted radiotherapeutics / radioconjugates) use **radioactive isotopes attached to drug molecules** to deliver radiation directly to cancer cells. It is **nuclear chemistry and pharmaceutical chemistry fused into one** — and Canada is a world leader, with 62 Health-Canada-approved radiopharmaceuticals and a startup engine producing billion-dollar companies. For Marcus, it collapses every fork in the road into a single, fundable, world-class path.

THE ROLE MODEL — STUDY THIS STORY CLOSELY

Dr. John Valliant was a chemistry professor at McMaster. He spun his research out of the university's radiopharma centre (the **CPDC**) into a company called **Fusion Pharmaceuticals** — building next-generation radioconjugates to treat cancer.

In June 2024, **AstraZeneca acquired Fusion for ~\$2.4 billion (USD)** — one of the largest acquisitions in Canadian university history. A chemist. A cancer therapy. A two-billion-dollar outcome. Built in Ontario, from a university lab.

And there's a reason this one should land for Marcus in particular: he is a cancer survivor. The single most fundable, meaningful thing a chemist can build right now is the kind of targeted therapy that saves people like him. That is not a metaphor — it is a market, a mission, and a career, all at once.

The people to meet

Named professors at Guelph, and the figures whose path he's following — start building these relationships in year one

Professors at Guelph (his actual department)

These are real, current Department of Chemistry faculty whose work maps to the pharma / drug-discovery / entrepreneurial path. The move: take their courses, go to office hours in first year, and ask to join a lab as early as possible. A professor's reference and a spot in their lab is the single highest-leverage relationship an undergrad can build.

PROFESSOR	WHY HE SHOULD KNOW THEM
Dr. Mario Monteiro	The entrepreneurial one to target first. Builds carbohydrate-based vaccines; his <i>Campylobacter jejuni</i> vaccine reached human trials and is WHO-listed. Living proof of a Guelph chemist turning lab work into real-world medicine.
Dr. Richard Manderville	Bioorganic / medicinal chemistry — DNA damage, aptamers, fluorescent probes. Core drug-discovery-adjacent science; strong research-lab mentor.
Dr. France-Isabelle Auzanneau	Carbohydrate & medicinal chemistry — synthesis of tumour-associated antigens (cancer-relevant). Premier's Research Excellence Award winner.
Dr. Derek O'Flaherty	Newer hire — nucleic-acid chemistry / RNA. Directly relevant to the oligonucleotide-therapeutics wave (the tech behind mRNA and RNA drugs).
Dr. Adrian Schwan · Dr. Rui Huang	Organic synthesis & methodology — the craft skills behind making any drug molecule. Good for mastering the bench fundamentals.

Tactic: he doesn't need to pick now. In first year, take their courses, talk to them, and let curiosity choose the lab. By second year, be in one.

The figure whose path he's walking

- ▶ **Dr. John Valliant (Fusion Pharmaceuticals / McMaster)** — the scientist-founder blueprint. Read every interview. If Marcus leans radiopharma, McMaster's ecosystem (an hour away) is where this path is most developed in Canada.
- ▶ **The CPDC network (McMaster)** — the radiopharma startup factory that launched Canprobe, ARTMS, Fusion, and AtomVie. A target for graduate study, internships, or eventual collaboration.

The companies — the whole map

Where a Guelph pharmaceutical chemist actually works — co-op, first job, or eventual competitor

TIER	COMPANIES (NAMED)	WHY THEY MATTER
Big Pharma (Ontario)	AstraZeneca, GSK, Roche, Amgen, Teva, Sanofi, Bayer, Apotex, Pharmascience	~50 pharma HQs in Toronto. Stable co-ops, brand-name résumé signal.
Drug-discovery CROs	Dalriada Drug Discovery (Ontario's largest), Sygnature Discovery, Charles River Labs	Where you learn real medicinal chemistry fast, across many projects.
CDMOs / manufacturing	Catalent, Dalton Pharma Services, ITR Laboratories, Applied Pharmaceutical Innovation	Process & analytical chemistry — the "make it at scale" skill set.
Radiopharma HIS FRONTIER	Fusion (now AstraZeneca), ARTMS, AtomVie, Canprobe, CPDC; Laurentis Energy Partners (OPG isotopes)	The nuclear-meets-pharma niche. Scarce talent = huge leverage.
Biotech startups	Deep Genomics, BlueRock Therapeutics, ProteinQure, BenchSci	The frontier of how drugs get discovered (AI, cell therapy, protein design).
The cluster / front doors	MaRS Discovery District, JLABS @ Toronto (J&J incubator), CCRM, OmniaBio	The physical hubs where founders, investors, and talent meet. Go to events.

HOW TO USE THIS MAP

Co-ops 1–2: get inside a **CRO or big-pharma lab** to build credible bench skills. Co-ops 3–4: aim at the **radiopharma niche** (Fusion/AstraZeneca, AtomVie, CPDC, Laurentis) where his nuclear interest becomes a rare, prized differentiator. Spend his elective energy and his network at **MaRS / JLABS events** — that's where the founder path begins.

The operator's timeline

Stage by stage — who to talk to, what to join, what to build, at each point

STAGE	DO THIS	WHO TO MEET / WHAT TO JOIN
Year 1	Clear the GPA gate. Take chemistry profs' courses; go to office hours. Read the Valliant/Fusion story. Get a free student membership in the Chemical Institute of Canada (CIC) .	Guelph chem professors (Monteiro, Manderville, Auzanneau). The Chemistry Club. A first campus entrepreneurship event at the John F. Wood Centre .
Year 2	Get <i>into a research lab</i> . Aim WT1 at a CRO or pharma QC role. Apply for an NSERC USRA summer research award.	Your lab supervisor (your most important reference). Co-op coordinator. Canadian Society for Chemistry (CSC) conference — go even as a spectator.
Year 3	Aim WT2/WT3 at the radiopharma niche (Fusion/AstraZeneca, AtomVie, CPDC, Laurentis). Do IgniteLab at the Wood Centre with any idea — just to learn the muscle.	Radiopharma scientists. Hub Incubator mentors. NAYGN (young nuclear professionals). First MaRS / JLABS event.
Year 4–5	Own a named research deliverable. Consider the CHEM*4900 research thesis if a startup idea or grad school is forming. Convert a co-op into a full-time offer.	A graduate supervisor (Guelph or McMaster/CPDC). LaunchLab pitch competition. Angel/founder contacts from MaRS.
Post-grad	Either: master's at a radiopharma lab (McMaster/CPDC) to deepen the niche → found from there; or industry role → build the network and idea, then jump.	Lab2Market → Creative Destruction Lab . Mitacs. An NRC-IRAP advisor. First angel investors.

How a scientist becomes a funded founder

The actual machinery — the programs and the money, in the order you use them

Nobody teaches science students this, so here it is plainly. You do **not** need to be rich or know investors. Canada has a deliberate ladder that takes a researcher with an idea and walks them to funding — and nearly all of it is non-dilutive (you keep your company).

Step 1 — Learn the muscle (at Guelph, while in school)

- ▶ **John F. Wood Centre** → **IgniteLab** (10-week fall program) → **LaunchLab** (+ pitch expo). Up to **\$5,000** and mentorship. Low stakes — the point is reps, not the idea.

Step 2 — Turn research into a company (after a degree/thesis)

- ▶ **Lab2Market** — the national program built exactly for this: *Discover* → *Validate* → *Launch*, with **\$15,000 Mitacs grants** at the later stages. Its graduates feed directly into Creative Destruction Lab.
- ▶ **Creative Destruction Lab (CDL)** — the elite science-startup accelerator; direct exposure to serious investors.
- ▶ **Velocity (Waterloo)** and **Next Canada** — alternative top-tier accelerators.

Step 3 — Stack the funding (non-dilutive first)

SOURCE	WHAT IT GIVES
Mitacs Accelerate	Subsidized grad-student/postdoc researchers (~\$15K units) — cheap R&D horsepower. Easiest entry point.
NRC IRAP	Up to \$1M for R&D salaries; a dedicated advisor; rolling intake, no deadline. The workhorse grant.
Innovative Solutions Canada	Phase 1 up to \$150K to prove the concept; Phase 2 up to \$1M to build the prototype.
NSERC Alliance	2:1 government match on research done with a university professor — keeps the lab relationship paying off.
SR&ED tax credits	Refunds a large share of R&D spend. Stack it on top of all the above (standard practice).

~\$5 billion a year flows through these programs, and they accept pre-revenue companies. The discipline is simply knowing they exist and applying — most founders don't until years in.

THE RADIOPHARMA FOUNDER'S SHORTCUT

If he goes the radiopharma route, the path is unusually paved: the **CPDC at McMaster** exists specifically to spin chemists' research into companies (it built Fusion). A master's or PhD there, plugged into that machine, is arguably the single highest-probability way in Canada for a chemist to become a venture-backed founder. **Valliant did exactly this.** The road map already exists — Marcus would just be the next to walk it.

The first moves — this year

Small, concrete, and entirely doable as a first-year student

- ▶ **Join the Chemical Institute of Canada** (free for students) — instant access to the professional community.
- ▶ **Read the Fusion / John Valliant story end to end** — let him see, concretely, that a chemist did this in Ontario.
- ▶ **Introduce himself to one professor** (Monteiro is the natural first) in office hours — not for anything, just to start the relationship.
- ▶ **Go to one John F. Wood Centre event** — see the entrepreneurial side of campus exists.
- ▶ **Follow MaRS, JLABS @ Toronto, and CPDC** — start absorbing the world he's aiming at.
- ▶ **Keep the floor first:** none of this matters if the GPA gate slips. Year one, the boss fight is still the grades. This is the horizon to lift his eyes toward — not a reason to drop the guard.

THE WHOLE DOCUMENT IN ONE LINE

He is not "a chemistry student deciding between jobs." He is one well-aimed degree away from the **frontier where nuclear, pharmaceutical chemistry, entrepreneurship, and curing cancer are the same field** — a field with a proven Canadian path from a university lab to a two-billion-dollar company. Point him at it, introduce him to the people, and let him build.

SOURCES & VERIFICATION · University of Guelph Department of Chemistry faculty (Monteiro — carbohydrate vaccines, C. jejuni human trials, WHO-listed; Manderville, Auzanneau, Schwan, Huang, O'Flaherty). Radiopharmaceuticals: AstraZeneca acquisition of Fusion Pharmaceuticals (~\$2.4B USD, completed June 2024); CPDC (McMaster) spinouts Canprobe, ARTMS, Fusion, AtomVie; founder John Valliant. Employers/cluster: MaRS Discovery District, JLABS @ Toronto, CCRM, OmniaBio, Dalriada Drug Discovery, Sygnature, Catalent, Dalton Pharma Services, ITR Labs; ~50 pharma HQs in Toronto. Entrepreneurship: U of G John F. Wood Centre (Hub Incubator / IgniteLab / LaunchLab, up to \$5K); Lab2Market (Mitacs, \$15K grants); Creative Destruction Lab; Velocity; Next Canada. Funding: NRC IRAP (to \$1M), Mitacs Accelerate, Innovative Solutions Canada (Phase 1 \$150K / Phase 2 \$1M), NSERC Alliance, SR&ED. Named professors/companies are research-and-relationship targets, not endorsements; confirm current faculty, programs, and openings directly. Prepared June 2026.